

Jordan Normal Form Cheat-Sheet

We're going to discuss only square matrices, so I will say "matrix" to implicitly mean "square matrix".

The *Jordan Normal Form* (JNF) of a matrix A is a specific almost-diagonal matrix J that is similar to A . When A is diagonalizable, J is the diagonalization of A . Otherwise, the matrix J is block-diagonal, and each block will be an almost-diagonal matrix of the form

$$\begin{pmatrix} \lambda & 1 & & & \\ & \lambda & 1 & & \\ & & \ddots & \ddots & \\ & & & \ddots & 1 \\ & & & & \lambda \end{pmatrix},$$

where λ is an eigenvalue of A . Such a block is called a *Jordan block*. In the JNF of A , the sum of the sizes of the Jordan blocks associated to λ will equal the algebraic multiplicity of λ as an eigenvalue of A . (e.g. if λ has algebraic multiplicity 3, then A could have one 3×3 Jordan block, a 2×2 Jordan block and a 1×1 Jordan block, or three 1×1 Jordan blocks.)

For example, the following 6×6 matrix has 3 Jordan blocks, one of size 1 and eigenvalue 1, one of size 2 and eigenvalue 1, and one of size 3 and eigenvalue 4:

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 1 & 0 \\ 0 & 0 & 0 & 0 & 4 & 1 \\ 0 & 0 & 0 & 0 & 0 & 4 \end{pmatrix}.$$

Saying that A is similar to J means that there is an invertible matrix B so that $A = BJB^{-1}$. The columns of the matrix B form a special basis for the underlying vector space called a *Jordan basis*. A Jordan basis is a union of *Jordan chains*, which is a sequence of vectors $v_k, v_{k-1}, \dots, v_1$ such that $Av_k = \lambda v_k + v_{k-1}$. This is what gives the Jordan blocks their staircase shape. In the example above, the Jordan chains making up the basis are $\{e_1\}$, $\{e_3, e_2\}$, and $\{e_6, e_5, e_4\}$.

The vectors v_i are examples of *generalized eigenvectors*: a generalized eigenvector of eigenvalue λ is an element of $\ker(A - \lambda I)^k$ for some natural number k ; the least such k I call the "degree" of the generalized eigenvector, while Wikipedia calls it the "rank". (I do not know whether anybody else uses my terminology, but it seems obviously bad in context to call this the "rank" of the generalized eigenvector.) You can only find generalized eigenvectors for λ that are already eigenvalues of A , because otherwise $A - \lambda I$ is invertible.

If you were given a matrix A and wanted to compute a Jordan basis for it, you would:

- i. Find the eigenvalues of A .
- ii. Choose an eigenvalue λ and compute the maximal generalized eigenspace $V_\lambda = \ker(A - \lambda I)^n$.

Choose almost any vector $v_k \in V_\lambda$, and start forming a Jordan chain by recursively defining $v_{i-1} = Av_i - \lambda v_i = (A - \lambda I)v_i$. (I say “almost any” because you need v_k to have maximal degree, and the generalized eigenvectors of lower degree form a measure 0 set.)

- iii. The Jordan chain you get in Step ii. might not span all of V_λ . If it doesn't, choose almost any vector in V_λ that isn't in the span of the vectors you already have and repeat the process. Continuing in this fashion, you will eventually have enough Jordan chains to span V_λ .
- iv. Repeat Steps ii. and iii. for all the other eigenvalues.

The JNF of a matrix is related to its characteristic polynomial. The characteristic polynomial of A is the product of $(x - \lambda_i)$ for each time you see λ_i on the diagonal of its JNF. For the example above, the characteristic polynomial is $(x - 1)^3(x - 4)^3$.

There is a different polynomial which is more closely associated to the JNF. The *minimal polynomial* of A is the lowest-degree polynomial A satisfies (meaning $f(A) = 0$). The minimal polynomial divides any other polynomial A satisfies. By the Cayley-Hamilton Theorem, A satisfies its own characteristic polynomial, so the minimal polynomial always divides the characteristic polynomial. The minimal polynomial of a $k \times k$ Jordan block of eigenvalue λ is the same as its characteristic polynomial: $(x - \lambda)^k$. However, the minimal polynomial of A is the gcd, not the product, of the minimal polynomials of its Jordan blocks. That means that in the minimal polynomial for A , $(x - \lambda_i)$ appears to the power k which is the size of the largest Jordan block. So, in the example above, A has minimal polynomial $(x - 1)^2(x - 4)^3$. (Notice: the minimal polynomial doesn't just divide the characteristic polynomial, but they also have all the same irreducible factors, and differ only in the multiplicity of each root.)

A Jordan normal form for a matrix always exists over $\mathbb{C}$, but might not exist over a field k which is not algebraically closed. It will exist if and only if all the eigenvalues lie in k . If you're over a non-algebraically closed k and have eigenvalues that do not lie in k (e.g. you have a real matrix with complex eigenvalues), there is another normal form called the *Rational Canonical Form* (RCF), which is also a block diagonal normal form. If the eigenvalue λ does not lie in k , then, instead of having a Jordan block, the RCF will have a *companion matrix* associated to a polynomial $f(x) \in k[x]$ such that $f(\lambda) = 0$. The companion matrix associated to $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0$ is the $n \times n$ matrix

$$\begin{pmatrix} 0 & 0 & \dots & \dots & -a_0 \\ 1 & 0 & & & \vdots \\ 0 & 1 & \ddots & & \vdots \\ \vdots & & \ddots & 0 & -a_{n-2} \\ 0 & \dots & & 1 & -a_{n-1} \end{pmatrix}.$$

The RCF of a matrix can be computed algorithmically, but it's more complicated than computing the JNF, and not worth going into. It's better to understand the RCF from the perspective of module theory, rather than trying to think about it purely in terms of linear algebra.